Théo **Schifferli**

Machine Learning Scientist at BIIE.AI

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Education

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

M.Sc. in Data Science – Computational Biology Minor — *Feb. 2024 - July 2026*

- Thesis on CAPO: Compute-Aware Automated Protein-Model Optimization, supervised by T. Bikias and M.-A. Hartley. Grade: 6/6.
- Presented work on Explainable AI in Personalised Cancer Medicine at IEEE SSCI 2025, awarded 2nd place in the competition.
- Notable projects:
  - Multimeditron: Developed at LiGHT Lab, focusing on a modular multimodal LLM trained from scratch with MoE and embedding-projection fusion.
  - Logprob-reward GRPO: Explored confidence-based rewards to elicit LLM reasoning across math and poetry tasks.

B.Sc. in Computer Science — *Sep. 2020 – Jan. 2024*

- Thesis on Multimodal Large Language Models for food recognition, supervised by Prof. Marcel Salathé. Grade: 6/6.

Work Experience

Machine Learning Scientist

Lausanne, Switzerland

Botnar Institute of Immune Engineering (BIIE) — *Oct. 2026 – present*

- Researching novel architectures and agentic systems, with efficient training and rigorous evaluation.

Master's Thesis Researcher

Lausanne, Switzerland

Botnar Institute of Immune Engineering, AI Initiative — *Jan. 2026 – Jul. 2026*

- Developed CAPO, a compute-aware agentic system for end-to-end protein-model optimization, with evaluation and ablations across protein-engineering tasks. Accepted at the NeurIPS 2026 AgenticLS workshop; won Best Poster Award at the OxML 2026 Health & Bio Track.

Research Intern

Lausanne, Switzerland

Botnar Institute of Immune Engineering, AI Initiative — *Sep. 2025 – Dec. 2025 (20%)*

- Built a GraphRAG pipeline (entity graph, Personalized PageRank) over the immunology corpus, benchmarked against production RAG.

Student Assistant in Modern NLP

Lausanne, Switzerland

EPFL — *Feb. 2025 - Jul. 2025*

- Mentored master's students in weekly labs and supervised and graded 12 final projects for CS-552: Modern NLP.

Summer Intern, Data Intelligence & Digitalisation

Geneva, Switzerland

Union Bancaire Privée (UBP) — *Aug. 2024 – Sep. 2024*

- Built Python, Flask and R Shiny tools to automate multi-timeframe trading-signal analysis and reporting.

Student Assistant in Theory of Computation

Lausanne, Switzerland

EPFL — *Feb. – Jul. 2023; Feb – Jul 2024*

- Supported students with course exercises during weekly two-hour sessions for CS-251: Theory of Computation.

Summer Research Intern, Regina Barzilay Group

Cambridge, MA, USA

MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) — *Aug. 2022 – Sep. 2022*

- Fine-tuned sequence models for pathogenicity prediction across 70 ClinVar cancer genes, achieving 0.875 AUC.

Extracurricular Activity

Not-So-Fast Run Club

Geneva, Switzerland

Co-Founder and President — *May 2024 – present*

- Co-founded and lead a non-profit Geneva running association with inclusive weekly runs, community events, partnerships with brands (including On, Saucony and Vitamin Well) and 1,000+ Strava members.

Technical Skills

Programming Python, C/C++, Java, Scala, C#, R, SQL, Assembly, VHDL

ML Stack PyTorch, NumPy, SciPy, Pandas, Jupyter, Hugging Face Transformers, Linux Command Line, Git

Bio/Protein Biopython, ESM, FASTA, AlphaFold, Boltz, ColabFold

Languages French (Native Speaker), English (Professional Proficiency), German (B2 Level), Spanish (B1 Level)

Interests Modern Art, Cooking, Long Distance Running, Music Production/Guitar, Skiing (Freestyle/Freeride), Triathlon